Project 7B

GIS Landslide Susceptibility Map of One Kerala District

Detailed guide to the project and the Frequency Ratio (FR) method

How to read this guide. Facts taken from your research draft (dated 28 Sep 2026) carry the draft's tags: **[V2]** = verified in 2+ sources, **[V1]** = one source, **[U]** = unverified. Parts marked **[Explainer]** are standard GIS and statistics practice that I am explaining; they are not from the draft's sources. All numbers in worked examples are **made up for illustration**.

1. Project at a glance

Item	Detail
Goal	Produce a map dividing one Kerala district into low / moderate / high / very high landslide susceptibility, and test how well it predicts landslides it has not seen.
Method	Frequency Ratio (FR) in QGIS with 6-8 conditioning factors, validated with ROC-AUC (from the draft).
Budget	28 hours: 8 weeks at about 3.5 h/week. The draft estimates about 6 h will go on learning QGIS.
Lab work	None. Everything uses free, open or registration-only data.
Role in draft	Backup to Project A (liquefaction, scored 24/25). Project B scored 18/25: open data 4, fits 28 h 3, professor-friendly 5, research gap 3, stays geotechnical 3.
Expected result quality	Published FR/AHP studies report AUC of about 0.82-0.92. Examples: Malappuram AHP 0.897 on test data; India-wide FR 0.905 [V1 each].

2. Why this topic is credible to a professor

- About 0.42 million sq km (12.6%) of India's land is landslide-prone (NDMA via Vikaspedia) **[V1]**. The draft suggests checking the NDMA NLRMP 2025 document for the same figure.
- The Landslide Atlas of India lists about 80,000 landslides between 1998 and 2022 and ranks 147 districts **[V2]**.
- Kerala districts rank high in the Atlas: Thrissur 3rd, Palakkad 5th, Malappuram 7th, Kozhikode 10th, Wayanad 13th **[V2]**.
- The Wayanad disaster of 30 July 2024 left over 400 people dead or missing. Sources differ: Wikipedia gives 420 deaths; Sci Rep 2025 gives 400 dead and 200 missing **[V2, range]**.

Plain-language pitch *[Explainer, wording is mine]*: Heavy monsoon rain on steep, weathered slopes can trigger landslides that destroy homes and roads. Using past landslide locations plus freely available terrain, rainfall and land-cover data, this project maps where in one district the same conditions occur, so planners know which areas deserve closer investigation.

3. Landslide susceptibility: the concept

[Explainer] Susceptibility answers the question *where* a landslide is likely, based on terrain and ground conditions. It is different from *hazard* (adds *when* and *how big*, i.e. rainfall return period and runout) and

from *risk* (adds what is exposed: people, buildings, roads). Your map is a susceptibility map only. Say this clearly in the report so it is not read as a hazard or risk map.

The usual assumption behind statistical methods like FR: **future landslides are more likely under the same conditions that produced past landslides.** FR turns this assumption into numbers by comparing, for each condition, where landslides occurred against how much of the district that condition covers.

4. The Frequency Ratio method

4.1 Definition

For one **class** of one **factor** (for example the class 30-45 degrees of the factor slope):

FR = (% of landslide points falling in the class) / (% of district area in the class)

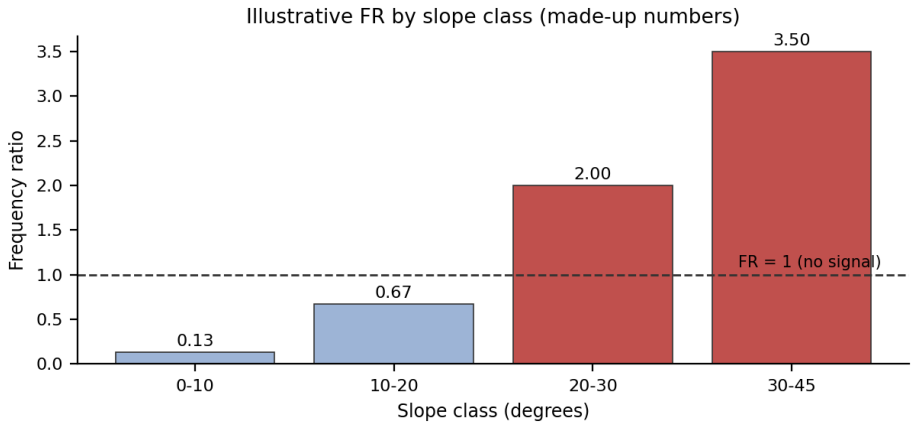
FR value	Meaning
FR = 1	Landslides occur in proportion to area. The class gives no information.
FR > 1	More landslides than area alone would predict. Class is landslide-prone.
FR < 1	Fewer landslides than expected. Class is relatively stable.
FR = 0	No training landslides in the class. Often caused by a very small class or a data gap (see section 7).

In plain terms, FR is a ratio of observed to expected landslides, where 'expected' is what you would see if landslides were scattered randomly over the district.

4.2 Worked example (illustrative numbers)

Suppose the training set has 200 landslide points. Slope is split into four classes:

Slope class	% of district area	Landslide points	% of landslides	FR
0-10 deg	40%	10	5%	5/40 = 0.13
10-20 deg	30%	40	20%	20/30 = 0.67
20-30 deg	20%	80	40%	40/20 = 2.00
30-45 deg	10%	70	35%	35/10 = 3.50
Total	100%	200	100%	



Steep slopes cover only 10% of the area but hold 35% of landslides, so FR = 3.5. Flat land covers 40% of the area but holds 5%, so FR = 0.13.

4.3 From class FR values to a susceptibility map

Each factor gets its own FR table. Every pixel in the district falls into exactly one class per factor, so you can look up an FR value for each factor and add them:

LSI (pixel) = FR(slope) + FR(aspect) + FR(curvature) + FR(lithology) + FR(land cover) + FR(rainfall) + FR(distance to stream) + FR(distance to road)

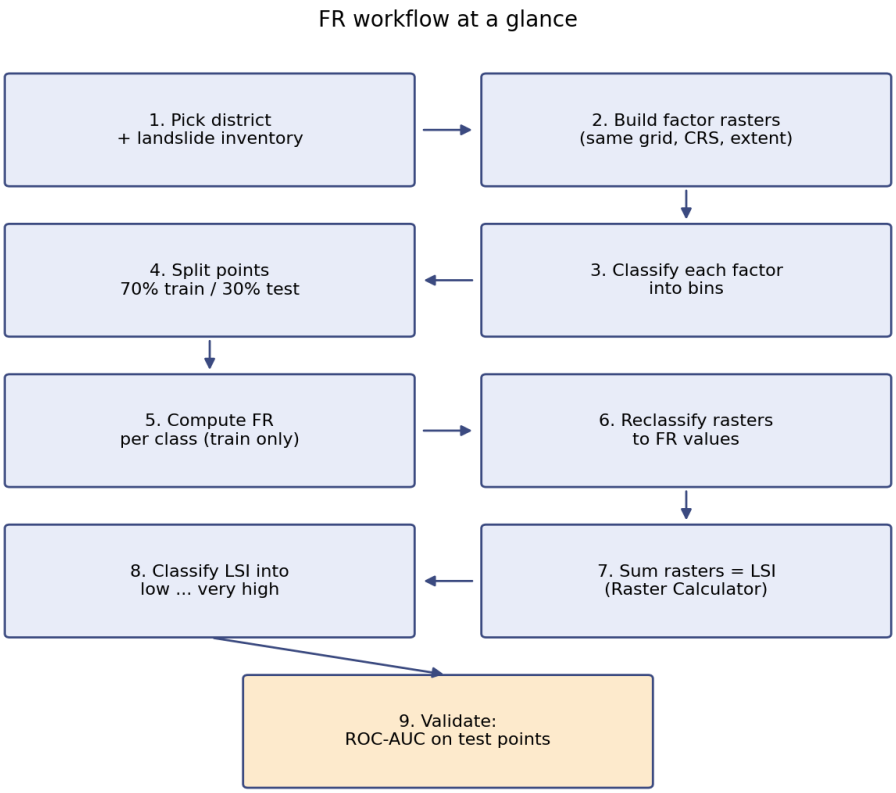
Example for one pixel (illustrative):

Factor	Class this pixel falls in	FR of that class
Slope	30-45 deg	3.50
Lithology	Class with weathered rock	1.80
Rainfall	Highest rainfall band	1.40
Land cover	Plantation	0.90
Distance to stream	100-200 m	1.20
Distance to road	> 500 m	0.70
LSI		**9.50**

Repeating this for every pixel gives a continuous LSI raster. A higher LSI means more of the pixel's conditions are ones where landslides were over-represented.

Note on weighting. Plain FR gives every factor equal weight. Some studies instead normalise each factor's FR values or weight factors using AHP or statistical models. The draft lists hybrid AHP+ML as future work in the literature; you can mention this without implementing it. *[Explainer]*

5. Step-by-step workflow



Step 1: Choose the district and get the landslide inventory

- Pick one district. Wayanad and Malappuram are the draft's examples. Small districts process faster; ensure the inventory has enough points (a rough rule of thumb is a few hundred; fewer makes many classes unstable). *[Explainer]*
- Inventory sources from the draft: GSI Bhukosh (registration needed) or the bharatlas mirror (30,842 points, CC0) **[V1]**.
- Clip points to the district boundary and count them. If the count is small, widen to a neighbouring district or merge with another source.
- Use **initiation points** (crown or source area) rather than whole runout polygons where possible, so the point represents the conditions where failure began. *[Explainer]*
- The draft notes that Kerala inventories are coarse and that an automated inventory is not available for the state (PMC12749052). Record the inventory's limitations in your report.

Step 2: Build the factor rasters

The draft lists 6-8 factors. Recommended for 28 hours: **6 factors**. All rasters must share one CRS (use a projected CRS such as UTM zone 43N or 44N for Kerala; check your district), one pixel size and one extent. *[Explainer]*

Factor	Source (from draft unless noted)	How to derive	Type
Slope	Copernicus GLO-30 DEM via OpenTopography (login needed)	QGIS: Raster > Analysis > Slope	Continuous
Aspect	Same DEM	QGIS: Aspect (treat flat as its own class)	Continuous (circular)

Factor	Source (from draft unless noted)	How to derive	Type
Curvature	Same DEM	SAGA or GRASS curvature tools	Continuous
Lithology	Draft lists the factor but no source. GSI Bhukosh geology layers are a candidate [U: check availability]	Rasterise the geology polygons	Categorical
Land cover	ESA WorldCover 10 m (no login)	Clip and resample	Categorical
Rainfall	CHIRPS	Mean annual or monsoon total; resample	Continuous
Distance to streams	Geofabrik OSM (waterways)	Proximity (raster distance)	Continuous
Distance to roads	Geofabrik OSM (roads)	Proximity (raster distance)	Continuous

Resolution mismatch *[Explainer]*: DEM is 30 m, WorldCover is 10 m, and CHIRPS is about 0.05 degrees (roughly 5 km). Resample everything to 30 m, but remember that a rainfall layer this coarse has only a handful of distinct values in one district, so it will separate areas weakly. State this as a limitation.

Step 3: Classify each factor into bins

- Categorical factors (lithology, land cover) already have classes.
- Continuous factors must be binned. Common choices: equal interval (e.g. slope in 5 or 10 degree steps), quantiles (equal area per class), or natural breaks (Jenks). Aim for roughly **5-10 classes** per factor. *[Explainer]*
- Aspect: use 8 compass directions plus 'flat'.
- Avoid classes so small that only a few landslide points can fall in them; their FR will be unstable.
- Keep your binning choices in a table. You will report them, and you can test sensitivity by changing them (section 8).

Step 4: Split the landslide points

- Randomly assign about 70% of points to **training** and 30% to **validation**. This 70/30 split is common in the literature. *[Explainer]*
- Compute FR using training points only. The validation points must stay unseen until Step 9.
- Generate non-landslide points (same number as validation landslides, randomly placed in areas away from any known landslide) for the ROC test.
- Record the random seed so your result can be reproduced.

Step 5: Compute FR for every class

In QGIS, use *Sample raster values* to attach each training point's class for every factor, and *Raster layer unique values report* (or zonal histogram) to get the pixel count per class. Export both to a spreadsheet and compute the ratio there; this is quick and transparent. Alternatively use Python:

```
import numpy as np

def frequency_ratio(class_raster, slide_mask, nodata=0):
    # class_raster: 2D int array of class IDs for one factor
    # slide_mask:   2D bool array, True where a TRAINING landslide point falls
    valid = class_raster != nodata
    n_pix = valid.sum()
    n_slides = (slide_mask & valid).sum()
    fr = {}
    for c in np.unique(class_raster[valid]):
        in_c = class_raster == c
        area_pct = in_c.sum() / n_pix
        slide_pct = (in_c & slide_mask).sum() / n_slides
        fr[int(c)] = slide_pct / area_pct
    return fr
```

Check: for each factor, the % of area column must total 100% and the % of landslides column must total 100%. If not, you have a nodata or alignment problem.

Step 6-7: Reclassify and sum

Replace each class ID in a factor raster by its FR value (QGIS: *Reclassify by table*). Then add all FR rasters with the Raster Calculator to get the Landslide Susceptibility Index (LSI).

Step 8: Classify the LSI map

Split the LSI into 4-5 zones. Common options are natural breaks, quantiles, or equal interval. Different methods give different-looking maps from the same LSI, so state which you used and why. A useful sanity check: the percentage of validation landslides per zone should rise from low to very high. *[Explainer]*

Step 9: Validate with ROC-AUC

[Explainer] Treat the LSI as a classifier. At every possible threshold, compute the true positive rate (validation landslides above the threshold) and the false positive rate (non-landslide points above the threshold). Plot true positive rate against false positive rate. The area under this ROC curve is the AUC.

AUC	Interpretation (rough convention)
0.5	No better than random
0.6-0.7	Poor to fair
0.7-0.8	Acceptable to good
0.8-0.9	Good to very good (published FR/AHP studies are about 0.82-0.92)
> 0.9	Excellent, but check for leakage or an easy non-landslide sample

- Report two AUCs: the **success rate** (training points, shows fit) and the **prediction rate** (validation points, shows predictive ability). The prediction rate is the one that matters.
- Python: `sklearn.metrics.roc_auc_score(y_true, lsi_values)` and `roc_curve`. QGIS also has plugins, but a short script is easier to document.

6. Quality checks before you trust the map

Check	What to do	Why
Factor correlation	Compute Pearson correlation or variance inflation factor (VIF) between factors. A common flag is |r| above 0.7 or VIF above 5-10.	FR adds factors as if independent; strongly correlated ones double-count.
Class sizes	Table of landslide count per class. Merge classes with very few points.	A class with 1-2 points produces a noisy FR.
Zero FR	Merge the class with a neighbour, or report it as 'no landslides in training set' rather than a hard zero.	A hard zero says a condition is impossible, which is overconfident.
Alignment	Overlay all rasters and landslide points visually; check same CRS and extent.	Misaligned layers silently give wrong class lookups.
Inventory bias	Plot points on a road map. Are they clustered along roads or settlements?	Easy-to-observe landslides are over-recorded and can inflate the 'distance to road' signal.
Sensible FR trends	Does FR rise with slope? Does it fall far from streams? Odd patterns need an explanation.	Reality check against mechanics.

7. Limitations to state in your report

- **Independence assumption.** FR is bivariate: each factor is analysed alone and then added.
- **Equal weights.** A weak factor (e.g. aspect) counts as much as a strong one (e.g. slope). AHP or ML methods can weight factors; the draft cites hybrid AHP+ML as suggested future work (PMC13534497).
- **Sensitivity to binning.** Different class boundaries give different FR values and a different map.
- **Inventory quality.** Coarse, incomplete, or roadside-biased inventories carry through to the map.
- **Random split optimism.** Landslides near each other share conditions, so a random split can give an optimistic AUC. A spatial split (train on one part of the district, test on another) is a stricter check. *[Explainer]*
- **Static map.** The map uses average rainfall, not a triggering storm, so it shows susceptibility, not hazard.
- **Resolution.** Coarse rainfall (about 5 km) and a 30 m DEM limit the detail of the result.

8. Optional extras if you have spare hours

- **Sensitivity test:** rerun with a different binning (e.g. quantiles vs equal interval) and compare AUC.
- **Factor-drop test:** remove one factor at a time and see how AUC changes. This shows which factors matter most.
- **Repeat the split** with 5 different random seeds and report the mean and range of AUC.
- **Comparison with a simple model** (logistic regression on the same factors) to show how FR performs against a multivariate method. This links to the draft's AHP+ML future-work gap.

9. Suggested 28-hour plan

[Explainer] The draft gives the liquefaction timeline only; this plan is my adaptation of the same 8-week, 3.5 h/week structure to 7B.

Week	Hours	Tasks	Deliverable
1	3.5	Choose district. Download inventory (register on Bhukosh early; use the bharatlas mirror as a fallback). Read 2 FR papers. Write 1-page proposal and get approval.	Proposal + inventory file
2	3.5	Install QGIS, learn basics (load, reproject, clip). Download the DEM, derive slope, aspect, curvature.	3 terrain rasters
3	3.5	Prepare land cover, rainfall, lithology, distance rasters. Align all to one grid.	6 aligned rasters
4	3.5	Bin factors, split points, sample values, compute FR tables in a spreadsheet.	FR tables
5	3.5	Reclassify, sum to LSI, classify into zones, make the map layout.	Susceptibility map
6	3.5	ROC-AUC (success and prediction rate), correlation check, optional sensitivity test.	ROC figure + checks
7	3.5	Write the report (outline in section 10).	Draft report
8	3.5	Buffer for slippage, then slides and viva preparation.	Final submission

Minimum viable version (if time runs short): Weeks 1-5 plus Week 7, with 5 factors and no extras. About 21 hours. *[Explainer]*

10. Suggested report outline

- **Introduction:** landslide problem in Kerala, Wayanad 2024, objectives.
- **Study area:** the district's geography, geology, rainfall, past events.
- **Data:** inventory, factor layers, sources, resolution, preprocessing.
- **Method:** FR definition, binning, train/test split, LSI, zoning, ROC-AUC.
- **Results:** FR tables and interpretation, susceptibility map, zone statistics, AUC (success and prediction rate).
- **Discussion:** which factors matter, comparison with published AUC values, inventory bias.
- **Limitations** (section 7) and **conclusions and future work** (hybrid AHP+ML, better inventory, rainfall-triggered hazard).
- **Appendix:** FR tables, binning table, code or QGIS model.

11. Claims you can and cannot make

You can say	You should not say
The Kerala landslide inventory is coarse and no automated inventory exists for the state (PMC12749052, per the draft).	'First Wayanad study.' ML studies of Wayanad already exist (HydroResearch 2025, per the draft).
Hybrid AHP+ML is suggested as future work in the literature (PMC13534497).	That your map is a hazard or risk map. It is susceptibility only.
Your AUC is comparable to or differs from published values of about 0.82-0.92, with reasons.	That a high AUC proves the map is correct everywhere. It only shows ranking performance on the test points.

12. Risks and mitigations

Risk	Mitigation
QGIS learning takes longer than the draft's 6 h estimate	Follow the QGIS Training Manual's Terrain Analysis lesson and a Frequency Ratio in QGIS tutorial (both listed in the draft). Do spreadsheet steps outside QGIS.
Portal logins or slow downloads (OpenTopography, Bhukosh)	Register in Week 1. Use the bharatlas mirror for the inventory.
Too few landslide points in the chosen district	Choose a district with more points, or combine neighbouring districts.
Professor sees the project as 'just GIS'	Emphasise geotechnical interpretation: slope angle, weathering and lithology, rainfall, drainage, and what the FR values imply about failure mechanisms. The draft scored this at 3/5.
Lithology data not found	Drop to 5 factors (the minimum viable version) and state why.

13. What to verify yourself

- The 12.6% landslide-prone figure is single-source [V1]; check the NDMA NLRMP 2025 document.
- Lithology data source: the draft does not name one. Check GSI Bhukosh and district geology maps.
- AUC figures from PMC13534497 and PMC12847720: read the papers to confirm and to copy their factor lists and binning.
- Bhukosh and bharatlas inventory formats and point counts for your chosen district.
- CHIRPS resolution and the best rainfall summary (mean annual vs monsoon) for a district-scale study.